

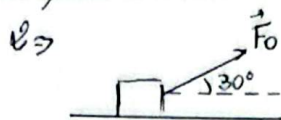
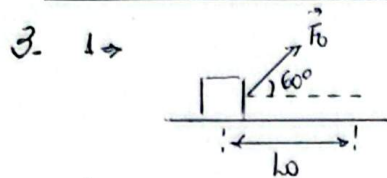
Componente Curricular: Física - Dinâmica

Professor(a): Rosana C. Pádua

Série: 2ª

Conteúdo: Trabalho - Exercícios para estudar - Bivros

Data: Março/Abul



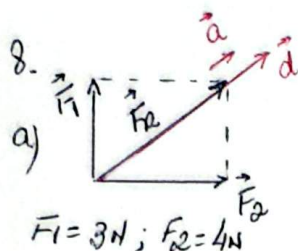
1 → $W_0 = F_0 \cdot l_0 \cdot \cos 60^\circ$

$W_0 = \frac{F_0 \cdot l_0}{2}$

2 → $W = F_0 \cdot l_0 \cdot \cos 30^\circ$

$W = \frac{\sqrt{3}}{2} \cdot F_0 \cdot l_0$

$\therefore W = \frac{\sqrt{3} \cdot F_0 \cdot l_0}{2} \Rightarrow |W = \sqrt{3} W_0|$



$F_R^2 = F_1^2 + F_2^2$

$F_R^2 = 3^2 + 4^2$

$F_R = \sqrt{25} \Rightarrow |F_R = 5N|$

$F_R = m \cdot a$

$5 = 2,5 \cdot a$

$|a = 2m/s^2|$

b) $\Delta S = \frac{a}{2} t^2$

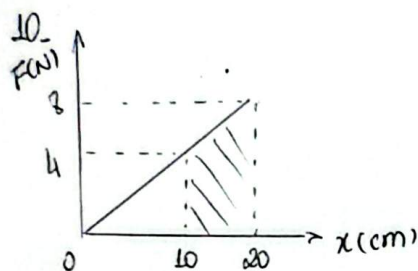
$\Delta S = \frac{2}{2} \cdot 3^2$

$\Delta S = 9m$

$\mathcal{E}_R = F_R \cdot d$

$\mathcal{E}_R = 5 \cdot 9$

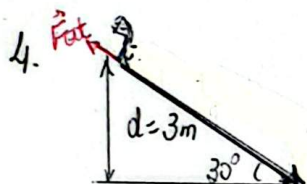
$|\mathcal{E}_R = 45J|$



$\mathcal{E}_{rel} = +A$ (for a liberada $\Rightarrow \mathcal{E} > 0$)

$\mathcal{E}_{rel} = \frac{(8+4) \cdot 0,1}{2}$ ($x = 10cm = 0,1m$)

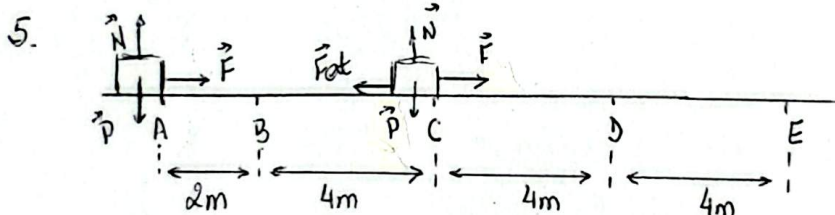
$\mathcal{E}_{rel} = 0,6J$



$\mathcal{E}_{fat} = -F_{at} \cdot d \Rightarrow \mathcal{E}_{fat} = -0,1 \cdot 3$

$\mathcal{E}_{fat} = -0,3J$

$\therefore |\mathcal{E}_{fat}| = 0,3J$



$F = 50N$

$F_{at} = 20N$

a) $\mathcal{E}_F = F \cdot d \Rightarrow \mathcal{E}_F = 50 \cdot 4 \Rightarrow \mathcal{E}_F = 200J$ (BC)

$\mathcal{E}_F = F \cdot d \Rightarrow \mathcal{E}_F = 50 \cdot 4 \Rightarrow \mathcal{E}_F = 200J$ (CD)

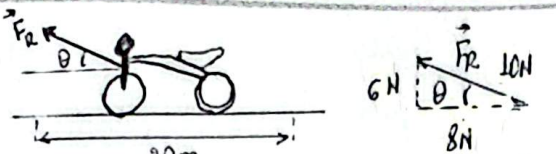
b) $\mathcal{E}_{fat} = -F_{at} \cdot d \Rightarrow \mathcal{E}_{fat} = -20 \cdot 4 \Rightarrow \mathcal{E}_{fat} = -80J$ (CD)

c) BC $\Rightarrow \mathcal{E}_R = \mathcal{E}_F \Rightarrow \mathcal{E}_R = 200J$

CD $\Rightarrow \mathcal{E}_R = \mathcal{E}_F + \mathcal{E}_{fat}$

$\mathcal{E}_R = 200 + (-80) \Rightarrow \mathcal{E}_R = 120J$

9.



$\vec{F}_R$
 θ
20m

6N
 $\vec{F}_R$ 10N
 θ
8N

$\cos \theta = \frac{8}{10} = \frac{4}{5}$

* $\mathcal{E}_R = F_R \cdot d \cdot \cos \theta$
 $\mathcal{E}_R = 10 \cdot 20 \cdot \frac{4}{5} \Rightarrow \mathcal{E}_R = 160 \text{ J}$

* Os lados do triângulo foram determinados pelo quadriculado